

Adding new user on multiple machines:

In this playbook I tested adding a new user on multiple machines. The user should be in sudo (wheel) group, the password should expire in 6 days and user should start receiving a warning about it 5 days before the expiration date. The username and password is taken from the encrypted files.

In ansible it's possible to use encrypted content for passwords stored in a file or in a variable. There is an option to have one vault password (recommended for smaller teams) or multiple passwords. Passwords can be memorized, stored in files or a third-party tool like system keyring or a secret manager. (Ansible Community Documentation, 2025)

In this playbook I created a single file and encrypted it with ansible vault following this tutorial (DigitalOcean, LLC., 2025)

- In **/home/ansibro/ansible_quickstart** created new directory called **vault** to store encrypted passwords.
- **ansible-vault create new_user.yml** and added password there for new users. Had to provide the password for the file hence I used the one I came up with.

And with **cat** command we can see encrypted file.

```
ansibro@alicedelice-Virtual-Machine:~/ansible_quickstart/vault$ ansible-vault create new_user.yml
New Vault password:
Confirm New Vault password:
ansibro@alicedelice-Virtual-Machine:~/ansible_quickstart/vault$ cat vault.yml
$ANSIBLE_VAULT;1.1;AES256
64336337326265336266643637366433376537663738313331393133623531353965623466323837
3232376232616464396435366131323533363964376539630a636364313764613465373237353933
63626639396138333666623830323933373236343065636330646634373138336665666232393765
3431333434666365370a336536393061653739343866653836666536383135313237313663376232
303639363962323623233613430383430616139633534346463343962383266333464
ansibro@alicedelice-Virtual-Machine:~/ansible_quickstart/vault$
```

- created also a file called **new_user_name.yml** that contains new user name since in this scenario, I will create only one user at the time but for 2 machines.

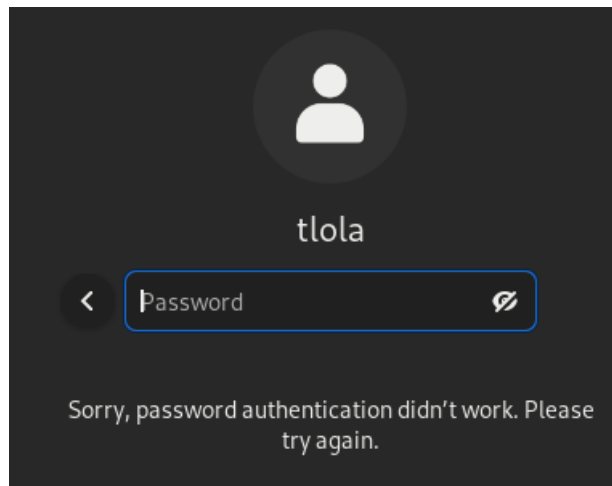
```
ansibro@alicedelice-Virtual-Machine:~/ansible_quickstart/vault$ ansible-vault create new_user_name.yml
New Vault password:
Confirm New Vault password:
ansibro@alicedelice-Virtual-Machine:~/ansible_quickstart/vault$ cat new_user_name.yml
$ANSIBLE_VAULT;1.1;AES256
64636262343838313939376432383961383638343231316264333965633962666630396330643837
3962613962356138333766353635316537666561373663610a373865663934346563313962303936
65663564353866626163336564656464363735386338616237336662613236636133346631616231
6134623231386134390a336437326163343339613131373730643033623031336131633736326261
3862
```

- in playbooks directory created a playbook called **new_users.yml**
- to create a new user I used a module for managing user accounts - **ansible.builtin.user module** (Ansible Community Documentation, 2025) where name parameter is required.
- If I try to run the playbook with just encrypted password, that was pulled out from vault, I'll get information **"[WARNING]: The input password appears not to have been hashed. The 'password' argument must be encrypted for this module to work properly."** when running the playbook

```
TASK [Create a new user] *****
[WARNING]: The input password appears not to have been hashed. The 'password' argument must be encrypted for this module to work properly.
changed: [Linux_Oracle_02]
changed: [Linux_Oracle_01]
```

```
password: "{ { user_passwd } }"
```

When trying to login, the following error appears:



It's because the system requires hashed password. I found on edureka community (edureka, 2020) that the playbook should look like this:

```
password: "{{ user_passwd | password_hash('sha512') }}"
```

- When trying to run the playbook though, I got the following error: “[ERROR]: Task failed: Finalization of task args for 'ansible.builtin.user' failed: Error while resolving value for 'password': The filter plugin 'ansible.builtin.password_hash' failed: Unable to encrypt nor hash, passlib must be installed: No module named 'passlib'”
- Used **sudo apt install python3-passlib** used to install the missing library on the control node however, ansible couldn't find it. I found by googling that issue may be caused if ansible was not installed with apt but as I remember I used pipx.

```
Task [create a new user] failed: [ERROR]: Task failed: Finalization of task args for 'ansible.builtin.user' failed: Error while resolving value for 'password': The filter plugin 'ansible.builtin.password_hash' failed: Unable to encrypt nor hash, passlib must be installed: No module named 'passlib'
Task failed.
Origin: /home/ansibro/ansible_quickstart/playbooks/new_users.yml:10:7
7
8
9 tasks:
10 - name: Create a new user
    column 7
```

- Used command **pipx inject ansible passlib** to install passlib.

```
ansibro@alicedelice-Virtual-Machine:~/ansible_quickstart/vault$ pipx inject ansible passlib
injected package passlib into venv ansible
done! 🌟 🌟 🌟
```

- Now, the playbook tasks were successful: **ansible-playbook --ask-become-pass --ask-vault-pass -i "/home/ansibro/ansible_quickstart/inventory/inventory.yaml" /home/ansibro/ansible_quickstart/playbooks/new_users.yml**
- **BECOME password:**
- **Vault password:**
- I can see I can log in to tlola account, I can use sudo command since tlola is in a wheel group and in /etc/passwd I can see that tlola has default shell /bin/bash and in /etc/shadow that tlola password is valid max 6 days and there will be a warning 5 days before.

```
[tlola@linux-2 ~]$ pwd
/home/tlola
[tlola@linux-2 ~]$ sudo su

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

[sudo] password for tlola:
[root@linux-2 tlola]# vim /etc/passwd
```

```
[tlola@linux-2 ~]$ sudo cat /etc/passwd | grep tlola
tlola:x:1003:1003::/home/tlola:/bin/bash
```

```
[tlola@linux-2 ~]$ groups
tlola wheel
```

```
[tlola@linux-2 ~]$ sudo cat /etc/shadow | grep tlola
tlola:$6$rounds=656000$VJxkaBfbeoZKP0b3$zEs5DUfUp9zli28LGHp5AYUYZfof4rp.F10TQdw6
u4EuD1XTUNGJZ1lvvFf8CGHlB7FeAj4o91OOVEpsL7xK7/:20412:0:6:5:::
```

Playbook:

```
---
- name: Create new user on myhosts
  hosts: myhosts
  become: true
  vars_files:
    - "/home/ansibro/ansible_quickstart/vault/new_user.yml"
    - "/home/ansibro/ansible_quickstart/vault/new_user_name.yml"

  tasks:
    - name: Create a new user
      ansible.builtin.user:
        name: "{{ username }}"
        password: "{{ user_passwd | password_hash('sha512') }}"
        groups: wheel
        password_expire_max: 6
        password_expire_warn: 5
        shell: /bin/bash
```

When logged in 3 days later to the created account, I can see the message that account will expire in 3 days. Everything works as expected.

```
tlola@linux-2:~  
login as: tlola  
tlola@192.168.100.4's password:  
Pre-authentication banner message from server:  
| Warning: your password will expire in 3 days.  
End of banner message from server  
Warning: your password will expire in 3 days.  
Activate the web console with: systemctl enable --now cockpit  
  
Last login: Thu Nov 20 15:05:30 2025 from 192.168.100.1  
[tlola@linux-2 ~]$
```

Bibliografia

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